

Full-result packet for arXiv:2406.03895

Finite atomicity exactly characterizes when normalized
lattice Lipschitz operators are strongly lattice Lipschitz

Status. full_solution_likely_valid. This packet answers the first open question on page 20 of Roger Arnau, Jose M. Calabuig, Ezgi Erdoğan, and Enrique A. Sánchez Pérez, *Lattice Lipschitz superposition operators on Banach function spaces*, arXiv:2406.03895v1. The source was subsequently published in the *Journal of Mathematical Analysis and Applications* 546(2) (2025), article 129233.

Source question. The source asks:

Which requirements are needed to assure that all lattice Lipschitz operators are strongly lattice Lipschitz? Which are the spaces—others than the trivial ones—for which this holds?

to search for the generalization of the above mentioned results in the context of the Lipschitz operators, for which it seems to be necessary to know the main properties of the particular class of superposition operators which we call lattice Lipschitz operators.

A lot of questions are still open regarding the results presented in this paper. Although we have shown some representation tools, we believe that it is possible to obtain more general results. We write below two open questions.

- Which requirements are needed to assure that all lattice Lipschitz operators are strongly lattice Lipschitz? Which are the spaces—others than the trivial ones—for which this holds?
- Are all the strongly lattice Lipschitz operators between Banach function spaces $X(\mu)$ and $Y(\mu)$ representable by strongly measurable functions in $\mathcal{M}(X, Y)$ with weaker requirements on $Y(\mu)$?

ETHICS DECLARATIONS

Conflict of interest. On behalf of all authors, the corresponding author states that there is no conflict of interest.

There is a normalization issue in the literal question. In the source, every strongly lattice Lipschitz map vanishes at zero, but a lattice Lipschitz map need not. Thus nonzero constant maps immediately rule out the literal inclusion. We resolve both the literal wording and its substantive normalized form.

Definitions. Let (Ω, Σ, μ) be a complete finite measure space, and let $X(\mu)$ and $Y(\mu)$ be Banach function spaces in the sense of the source; in particular $\mathbf{1} = \chi_\Omega$ belongs to both. A map $T : X \rightarrow Y$ is *lattice Lipschitz* if there is a finite, nonnegative real-valued function K such that, for every $f, g \in X$,

$$|T(f)(w) - T(g)(w)| \leq K(w)|f(w) - g(w)| \quad \text{for almost every } w.$$

No measurability of K is needed below in the positive direction; the bound used in the counterexample is measurable.

The map is *strongly lattice Lipschitz* if

$$T(f)(w) = \Phi(w)(f(w))$$

for a strongly measurable $\Phi : \Omega \rightarrow \text{Lip}_0(\mathbb{R})$ whose norm function $w \mapsto \text{Lip}(\Phi(w))$ belongs to the multiplier space $\mathcal{M}(X, Y)$. We write $\text{LLip}_0(X, Y)$ for the lattice Lipschitz maps satisfying $T(0) = 0$.

Proof intuition. There are two sharply opposed mechanisms. On finitely many atoms, every function is just a finite coordinate vector. The pointwise inequality makes each output coordinate depend only on its corresponding input coordinate, and its scalar coordinate function must be Lipschitz. A finite list of such functions is automatically strongly measurable, while every finite coordinate multiplier is bounded between X and Y .

If the measure algebra is infinite, it contains disjoint positive-measure sets A_n . Put increasingly large coefficients on these sets to obtain a finite-a.e. measurable function $K \notin Y$, and clip Kf to the interval $[-1, 1]$. Clipping guarantees that the output always lies in Y , while the exact pointwise Lipschitz constant remains K . Any strong representation is forced, already by constant rational inputs, to have norm function K ; but K is not a multiplier. This gives a normalized counterexample for every pair X, Y , independent of their finer geometry.

Lemma 1 (Measure-algebra dichotomy). *The measure algebra of a finite measure space is finite if and only if, modulo null sets, Ω is a union of finitely many atoms. If the measure algebra is infinite, there are pairwise disjoint sets $A_1, A_2, \dots \in \Sigma$ of positive measure.*

Proof. A finite Boolean algebra is generated by its finitely many atoms, giving the first assertion in one direction; the converse is immediate. For the second assertion, if there are infinitely many atoms, choose a countable subfamily. Otherwise remove the union of the finitely many atoms. If the remainder had measure zero, the measure algebra would be finite. Its positive remainder is therefore atomless, and repeatedly splitting a positive set into two positive parts produces the required disjoint sequence. $\square$

Theorem 2 (Complete normalized classification). *Fix Banach function spaces $X(\mu)$ and $Y(\mu)$ over the same complete finite measure space. The following are equivalent:*

1. *every $T \in \text{LLip}_0(X, Y)$ is strongly lattice Lipschitz;*
2. *the measure algebra of (Ω, Σ, μ) is finite;*
3. *modulo null sets, Ω is a union of finitely many atoms.*

Consequently, the answer depends only on the measure algebra and not on the particular Banach function norms.

Proof. The equivalence of (2) and (3) is Lemma 1.

Assume first that $\Omega = A_1 \sqcup \dots \sqcup A_N$ modulo null sets, where the A_i are atoms. Every measurable scalar function is constant almost everywhere on each A_i . Because $\mathbf{1} \in X \cap Y$ and both spaces are ideals, X and Y are, as vector spaces, both the full coordinate space $\mathbb{R}^N$, possibly with different lattice norms.

Take $T \in \text{LLip}_0(X, Y)$ with bound function K . For $i \leq N$ define $\varphi_i(t)$ to be the almost-everywhere constant value on A_i of $T(t\mathbf{1})$. Then $\varphi_i(0) = 0$. We claim that φ_i is Lipschitz. If it were not, for every n we could choose $s_n \neq t_n$ such that

$$\frac{|\varphi_i(t_n) - \varphi_i(s_n)|}{|t_n - s_n|} > n.$$

Apply the lattice Lipschitz inequality to $t_n\mathbf{1}$ and $s_n\mathbf{1}$. For almost every $w \in A_i$ it forces $K(w) > n$. After intersecting the countably many conull sets, one obtains a point $w \in A_i$ at which $K(w) > n$ for every n , contrary to K being finite-valued. Hence $L_i := \text{Lip}(\varphi_i) < \infty$.

Define the finite-valued field

$$\Phi(w) = \varphi_i \quad (w \in A_i).$$

It is strongly measurable and takes values in $\text{Lip}_0(\mathbb{R})$. Its norm function is the simple function $h = \sum_i L_i \chi_{A_i}$. Multiplication by h is a linear map from the finite-dimensional normed space $X = \mathbb{R}^N$ to the finite-dimensional normed space $Y = \mathbb{R}^N$, so $h \in \mathcal{M}(X, Y)$.

It remains only to identify the representation. If f has value f_i on A_i , then f and $f_i \mathbf{1}$ agree on A_i . The lattice Lipschitz inequality therefore says that $T(f)$ and $T(f_i \mathbf{1})$ agree on A_i . Thus

$$T(f)(w) = \varphi_i(f_i) = \Phi(w)(f(w)) \quad \text{for almost every } w \in A_i.$$

Hence T is strongly lattice Lipschitz, proving (3) $\Rightarrow$ (1).

Conversely, suppose the measure algebra is infinite. By Lemma 1, choose pairwise disjoint positive-measure sets A_n . Since χ_{A_n} is nonzero in Y , put

$$b_n = \frac{n}{\|\chi_{A_n}\|_Y}, \quad K = \sum_{n=1}^{\infty} b_n \chi_{A_n},$$

with $K = 0$ off their union. This K is measurable and finite everywhere. It does not belong to Y : otherwise the ideal property would give, for every n ,

$$\|K\|_Y \geq b_n \|\chi_{A_n}\|_Y = n.$$

In particular $K \notin \mathcal{M}(X, Y)$, since $\mathbf{1} \in X$ and a multiplier K would satisfy $K\mathbf{1} \in Y$.

Let

$$\theta(u) = \max\{-1, \min\{1, u\}\}, \quad T(f)(w) = \theta(K(w)f(w)).$$

Then $T(0) = 0$ and $|T(f)| \leq \mathbf{1}$, so $T(f) \in Y$ for every $f \in X$ by the ideal property. Since θ is 1-Lipschitz,

$$|T(f)(w) - T(g)(w)| \leq K(w)|f(w) - g(w)|,$$

and therefore $T \in \text{LLip}_0(X, Y)$.

Suppose, toward a contradiction, that T were strongly lattice Lipschitz, represented by some $\Phi : \Omega \rightarrow \text{Lip}_0(\mathbb{R})$. For every rational q , applying the representation to $q\mathbf{1}$ yields

$$\Phi(w)(q) = \theta(K(w)q)$$

outside a null set. Intersecting over the countable set of rationals and using continuity in the scalar variable gives, on one conull set,

$$\Phi(w)(t) = \theta(K(w)t) \quad (t \in \mathbb{R}).$$

The exact Lipschitz constant of $t \mapsto \theta(K(w)t)$ is $K(w)$: for $K(w) > 0$ this is attained on a sufficiently small interval around zero, and for $K(w) = 0$ both sides vanish. Thus $\text{Lip}(\Phi(\cdot)) = K$ almost everywhere. Strong lattice Lipschitzness would require this function to belong to $\mathcal{M}(X, Y)$, contradicting the construction of K . This disproves (1) whenever the measure algebra is infinite. $\square$

Corollary 3 (Literal and affine forms of the answer). *If $Y \neq \{0\}$, it is never true that every lattice Lipschitz map $T : X \rightarrow Y$ is strongly lattice Lipschitz: for $0 \neq y_0 \in Y$, the constant map $T(f) = y_0$ has bound zero but does not vanish at zero.*

The normalization-corrected affine statement

$$\text{every lattice Lipschitz } T : X \rightarrow Y \text{ has } T - T(0) \in \text{SLL}(X, Y)$$

holds if and only if the measure algebra is finite.

Proof. Only the affine assertion needs comment. Subtracting $T(0)$ preserves the lattice Lipschitz bound and produces a normalized map, so it follows from Theorem 2. The theorem's clipped counterexample is already normalized, proving necessity. $\square$

Verification notes. The negative example has no hidden measurability defect. Its natural field $w \mapsto [t \mapsto \theta(K(w)t)]$ has countable range and is the pointwise limit of finite simple Lip_0 -valued functions, hence is strongly measurable. The sole obstruction is exactly the one required by the source definition: its norm function is not in $\mathcal{M}(X, Y)$. No Fatou property, order-continuity, embedding $X \subseteq Y$, or special formula for $\mathcal{M}(X, Y)$ is used.

The only countable-null-set argument in the positive proof is applied to the explicit sequence of pairs witnessing an infinite scalar Lipschitz constant; there is no illegitimate intersection over all real inputs. Likewise, the uniqueness step in the negative proof uses only rational constant inputs and then continuity.

Novelty check and limitations. Before promotion, the run indexes were searched for arXiv:2406.03895 and for the core phrases “lattice Lipschitz”, “strongly lattice Lipschitz”, and the exact open-question sentence; no prior packet or duplicate was found. The current arXiv record is still v1 and still contains the question. Bounded web searches for the exact sentence, for “all lattice Lipschitz operators are strongly lattice Lipschitz”, and for the title plus later work found the source/published article and the authors’ 2025 paper on $C(K)$ spaces, but no stated resolution of this Banach-function-space question. The $C(K)$ paper assumes continuous functional bounds in a different category and does not supply the classification above. This is a bounded, not exhaustive, novelty check; mathematical validity and novelty both remain subject to expert review.

Human-review recommendation. Review as a short full classification. The most useful checks are: (i) the finite-atomic argument that a possibly nonmeasurable finite bound still forces each scalar coordinate map to have finite Lipschitz constant; (ii) the multiplier conclusion in finite dimension; and (iii) the rational-input uniqueness argument showing that clipping has exact norm field K .

References

- [1] R. Arnau, J. M. Calabuig, E. Erdoğan, and E. A. Sánchez Pérez, *Lattice Lipschitz superposition operators on Banach function spaces*, arXiv:2406.03895v1, 2024; J. Math. Anal. Appl. 546(2) (2025), 129233.
- [2] R. Arnau, J. M. Calabuig, and E. A. Sánchez-Pérez, *Lattice Lipschitz operators on $C(K)$ -spaces*, Ann. Funct. Anal. 16 (2025), article 32.